

1 CELL

THE NUMBER OF SPERM CELLS INJECTED DIRECTLY INTO AN EGG DURING AN INTRACYTOPLASMIC SPERM INJECTION

During intracytoplasmic sperm injection (ICSI), a single sperm cell is placed directly inside an ovum via an extremely fine glass needle.

ASTRONOMERS HAD ASSUMED for decades that there existed **extrasolar planets**—planets outside our Solar System. They had found several possible candidates, but **no definitive proof**. Confirmation came at last in 1992, with the detection of a planet orbiting a pulsar—which is a rotating neutron star that

significant development in cosmology since the discovery of cosmic background radiation (CMB, see 1964). COBE carried out an extremely sensitive all-sky survey of the CMB, and **detected slight variations in the radiation**, which correspond to slight variations in temperature in the early Universe. These

“ MERRY CHRISTMAS. ”

Neil Papworth, British engineer, first SMS, 1992

emits beams of radio waves. Within three years, astronomers had detected a planet in orbit around an ordinary star—one that is in the main part of its lifetime (see 1995).

Data collected by the **Cosmic Background Explorer** satellite (COBE) led to the most

temperature variations in turn represent variations in density. The density variations are important—if the density had been perfectly uniform, matter would never have clumped together to form stars, galaxies, and galaxy clusters.

In June 1992, representatives of governments and non-governmental organizations (NGOs) attended the **UN Conference on**

hole in ozone above Antarctica

Ozone hole
At the end of September 1992, atmospheric scientists reported that the ozone hole over the Antarctic (see 1985) had grown by 15 percent in a single year.

COSMIC BACKGROUND RADIATION

The cosmic background radiation (CMB) is the heat radiation that filled the Universe 380,000 years after the Big Bang (see pp.342–43), and it provides a record of the temperature of the Universe at that time. The CMB is remarkably isotropic—the same in every direction. Slight variations, or anisotropies, in the CMB are shown in the false-colour map above—red areas are slightly warmer than blue.

Environment and Development in Rio de Janeiro, Brazil, commonly referred to as the **Earth Summit**. The main goal of the conference was to discuss issues concerning the **sustainable use of Earth's natural resources** in an increasingly industrialized and highly populated world. Two major conventions came out of the summit. The first was the **Convention on Biological Diversity** in 1993 (see opposite); the second convention was the United Nations Framework Convention on Climate Change

(UNFCCC). An ambitious initiative aimed at combatting climate change—called the Kyoto Protocol (see 1997)—arose from the UNFCCC, and came into force in 2005.

At the end of September 1992, atmospheric scientists reported that the **ozone hole over the Antarctic** (see 1985) had grown by 15 percent in a single year—and was **the size of North America**.

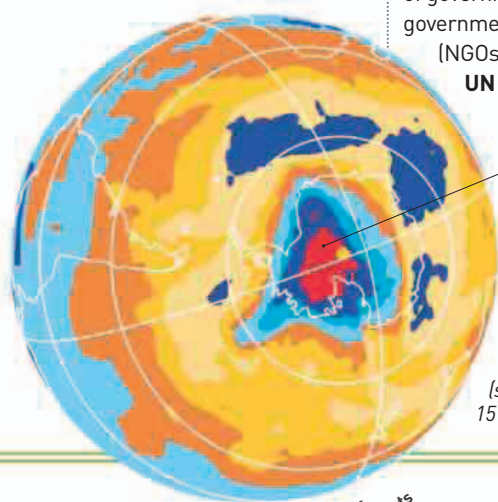
In vitro fertilization (see 1978) had been developed mainly to overcome female infertility; in the early 1990s, an extension of the technology—**intracytoplasmic**

sperm injection (ICSI)—was developed mainly to address male infertility. This revolutionary technique involves injecting a single sperm into an ovum (egg cell) to overcome problems arising from low sperm count or low sperm motility (ability to move toward the ovum). It was developed by Italian fertility specialist **Gianpiero Palermo** and Belgian doctor **André van Steirteghem** at the Vrije Universiteit Brussel in Belgium. The first births resulting from the technique were confirmed in 1992.

Small, simple **liquid crystal displays (LCDs)** had been available since the 1970s and were incorporated into consumer electronics devices such as calculators, digital watches, and video cassette recorders. In 1992, the

hydraulic suspension
carbon fiber lower leg

Prosthetic knee
This microprocessor-controlled knee prosthesis automatically adjusts to the wearer's gait.



January 9, 1992 Astronomers discover the first confirmed planets outside our Solar System

April 23, 1992 Scientists find the seeds of galaxy formation in data from the Cosmic Background Explorer (COBE)

June 3, 1992 The Earth Summit in Rio de Janeiro, Brazil, opens, bringing together 172 countries to discuss sustainability

September 29, 1992 NASA reports that the ozone hole over Antarctica grew rapidly during the year

1992 Hitachi develops the first practical, high-resolution LCDs (liquid crystal displays)

December 3, 1992 The first SMS message sent over a GSM network

1992 First birth of a child conceived by direct injection of sperm into an egg

1993 Blatchford releases the Intelligent Prosthesis, the first commercially available microprocessor-controlled prosthetic knee